

Quantum Algorithm for Computing the Period Lattice of an Infrastructure

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- ▶ Based on
 - ▶ *Quantum Algorithms for Computing the Period Lattice of an Infrastructure*
arXiv:1111.1348
 - ▶ *On the Probability of Generating a Lattice*
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 - ▶ *Computing Approximate Bases of a Lattice L and Its Dual L^* from an Approximate Generating Set of L*
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Quantum Algorithms and Hidden Lattice Problem over $\mathbb{Z}^n$

- ▶ Efficient quantum algorithms provide the motivation for building a quantum computer
- ▶ Many quantum algorithms that achieve an exponential speed-up over classical algorithms can be formulated as a hidden lattice problem over $\mathbb{Z}^n$
- ▶ The hidden lattice problem over $\mathbb{Z}^n$ is very closely related to the hidden subgroup problem

Definition of the Hidden Lattice Problem over $\mathbb{Z}^n$

- ▶ Let A be a finite abelian group with generators $a_1, \dots, a_n$
- ▶ Let $h : \mathbb{Z}^n \rightarrow A$ be a surjective group homomorphism given by

$$h(x_1, \dots, x_n) = a_1^{x_1} \cdots a_n^{x_n}$$

- ▶ We assume that we can evaluate h efficiently
- ▶ We have the isomorphism

$$A \cong \mathbb{Z}^n / \Lambda \quad \text{where} \quad \Lambda = \ker h$$

- ▶ Λ is a full-rank lattice in $\mathbb{Z}^n$
- ▶ The task is to determine a basis $\lambda_1, \dots, \lambda_n$ of $\mathbb{Z}^n$

$$\Lambda = \mathbb{Z}\lambda_1 + \dots + \mathbb{Z}\lambda_n$$

Integer Factorization as Hidden Lattice Problem

- ▶ Let n be the integer that we seek to factor
- ▶ Choose an element a uniformly at random from the unit group

$$\mathbb{Z}_n^\times = \{m \mid 1 \leq m < n, \gcd(m, n) = 1\}$$

- ▶ The surjective group homomorphism $h : \mathbb{Z} \rightarrow \mathbb{Z}_n^\times$ is given by

$$h(x) = a^x \pmod{n}$$

- ▶ The kernel of h is the lattice

$$\Lambda = r\mathbb{Z}$$

where r is the order of a , i.e., the smallest positive integer r such that $a^r = 1 \pmod{n}$

- ▶ With high probability r is even and $\gcd(a^{r/2} - 1, n)$ yields a non-trivial factor

Discrete Logarithm Problem as Hidden Lattice Problem

- ▶ Let a be a fixed generator of the unit group $\mathbb{F}_q^\times$
- ▶ Let b a given element of $\mathbb{F}_q^\times$
- ▶ The task is to determine the discrete logarithm of b with respect to the generator a , i.e., an integer $r \in \{0, \dots, q-2\}$ such that

$$a^r = b$$

- ▶ The surjective homomorphism $h : \mathbb{Z}^2 \rightarrow \mathbb{F}_q^\times$ given by

$$h(x, y) = b^x \cdot a^y$$

- ▶ The kernel of f is the lattice Λ generated by

$$\Lambda = \mathbb{Z} \begin{pmatrix} 0 \\ q-1 \end{pmatrix} + \mathbb{Z} \begin{pmatrix} 1 \\ -r \end{pmatrix}$$

Quantum Algorithms and Hidden Lattice Problem over $\mathbb{R}^n$

- ▶ The quantum algorithms due to HALLGREN and VOLLMER AND SCHMIDT for the number theoretic problems
 - ▶ solving Pell's equation
 - ▶ computing the unit group
 - ▶ solving the principal ideal problem
 - ▶ computing the class group

can all be formulated as a hidden lattice problem over $\mathbb{R}^n$

- ▶ These running times of these quantum algorithms scales
 - ▶ **polynomially** in the logarithm of the discriminant $\Delta_{K/\mathbb{Q}}$ of the number field K ,
 - ▶ but **exponentially** in $q(n)$ where n is the unit rank of K and $q : \mathbb{N} \rightarrow \mathbb{N}$ is a polynomial

Improvements

- ▶ Achieve conceptual simplification by encapsulating all number theoretic details using infrastructures
- ▶ Provide a mathematically rigorous and complete analysis of the quantum algorithm
- ▶ Improve the running time by an exponential factor

n -Dimensional Tori

- ▶ Let Λ be a full-rank lattice in $\mathbb{R}^n$, i.e.,

$$\Lambda = \mathbb{Z}\lambda_1 + \dots + \mathbb{Z}\lambda_n$$

where $\lambda_1, \dots, \lambda_n$ are linearly independent vectors in $\mathbb{R}^n$

- ▶ The abelian group $T = \mathbb{R}^n/\Lambda$ is an n -dimensional torus
- ▶ A one-dimensional torus $T = \mathbb{R}/(\mathbb{Z}\lambda)$ is simply a circle of circumference λ

Hidden Lattice Problem over $\mathbb{R}^n$

- ▶ Let $h : \mathbb{R}^n \rightarrow A$ be a surjective group homomorphism such that

$$\Lambda = \ker f$$

is a full-rank lattice in $\mathbb{R}^n$, i.e., A is isomorphic to the n -dimensional torus

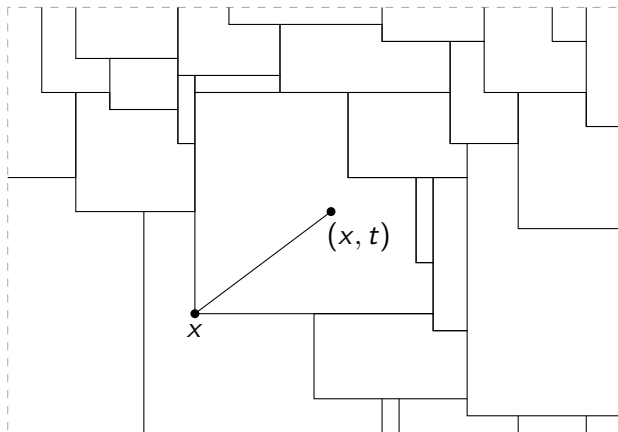
$$T = \mathbb{R}^n / \Lambda$$

- ▶ Assume that h can be evaluated “approximately”
- ▶ The task is to find a γ -approximate basis of $\tilde{\lambda}_1, \dots, \tilde{\lambda}_n$ of Λ

$$\|\lambda_i - \tilde{\lambda}_i\|_2 \leq \gamma \quad \text{for } i = 1, \dots, n$$

where $\lambda_1, \dots, \lambda_n$ is some basis of Λ

Abstract Infrastructures – Intuition



- ▶ The regions are “cornered”
- ▶ The lower-left corners of the regions are labeled by elements x of some finite set X
- ▶ Every point is uniquely identified by a pair (x, t)

Abstract Infrastructures – Formal Definition

An n -dimensional infrastructure $\mathcal{I}$ consists of

- ▶ a full-rank lattice Λ in $\mathbb{R}^n$ (called the period lattice)
- ▶ a finite non-empty set X
- ▶ an injective map $d : X \rightarrow T$ where $T = \mathbb{R}^n / \Lambda$
- ▶ a set of f -representations $\text{Rep}(\mathcal{I}) \subseteq X \times \mathbb{R}^n$ such that

$$\Phi_{\mathcal{I}} : \text{Rep}(\mathcal{I}) \rightarrow \mathbb{R}^n / \Lambda, \quad (x, t) \mapsto d(x) + (t \bmod \Lambda)$$

is a bijection

Abstract Infrastructures – Group Structure

- ▶ $\text{Rep}(\mathcal{I})$ is endowed with an abelian structure:

$$(x, t) + (x', t') := \Phi_{\mathcal{I}}^{-1}(\Phi_{\mathcal{I}}(x, t) + \Phi_{\mathcal{I}}(x', t'))$$

- ▶ There is a surjective group homomorphism h given by

$$h : \mathbb{R}^n \rightarrow \text{Rep}(\mathcal{I}), \quad v \mapsto \Phi_{\mathcal{I}}^{-1}(v \bmod \Lambda)$$

Concrete Infrastructures Arise From Number Fields

- ▶ Let K be a number field of degree $d = [K : \mathbb{Q}]$ and unit rank $n = r + s - 1$ where r is the number of real embeddings and $2s$ the number of complex embeddings of K
- ▶ The lattice Λ is the kernel of the logarithmic map in Dirichlet's Unit theorem

This theorem gives the structure of the unit group $\mathcal{O}_K^\times$ of the ring of integers $\mathcal{O}_K$ of K

- ▶ The set X is the set of reduced fractional ideals of $\mathcal{O}_K$

Perfect Periodicity is Impossible

- ▶ The task is to determine an approximate basis of the period lattice Λ hidden by the surjective group homomorphism

$$h : \mathbb{R}^n \rightarrow \text{Rep}(\mathcal{I}) \subset X \times \mathbb{R}^n \quad \text{with} \quad \ker f = \Lambda$$

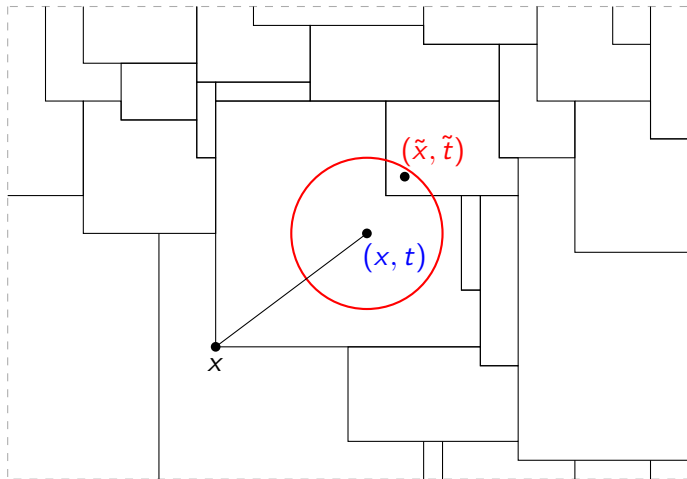
- ▶ But there are two fundamental problems:
 - ▶ We can only evaluate h at a finite number of points $v \in \mathbb{R}^n$
 - ▶ We cannot compute the correct value $h(v) = (x, t)$

We only obtain an approximate f -representation $\tilde{h}(v) = (\tilde{x}, \tilde{t})$ such that

$$\Phi_{\mathcal{I}}(x, t) = \Phi_{\mathcal{I}}(\tilde{x}, \tilde{t}) + (\delta \bmod \Lambda)$$

where the norm $\|\delta\|$ of the deviation $\delta \in \mathbb{R}^n$ is small

Approximate f -representations



- The **circle** has radius $\|\delta\|$

Pseudo-Periodicity is Possible

- ▶ Let
 - ▶ q and N be sufficiently large positive integers
 - ▶ $\mathcal{V} = \{0, \dots, qN - 1\}^n$
 - ▶ L an integer sufficiently greater than N
 - ▶ $s \in \{1, \dots, L\}$ is chosen uniformly at random
- ▶ Evaluate only at points of the form

$$\frac{s}{L} + \frac{v}{N} \quad \text{for } v \in \mathcal{V}$$

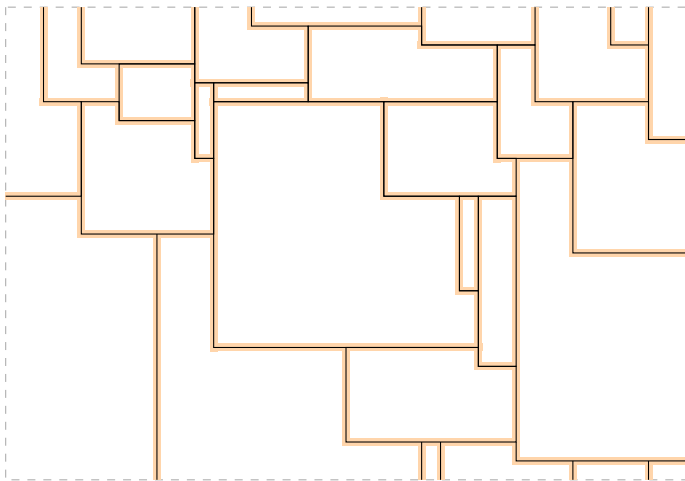
- ▶ With constant probability, all evaluation points avoid certain **forbidden regions** so that

$$(x, \lfloor t/N \rfloor) = (\tilde{x}, \lfloor \tilde{t}/N \rfloor)$$

provided that **the approximation error** $\|\delta\|$ is smaller than $\frac{1}{2}$ of the width of the forbidden regions

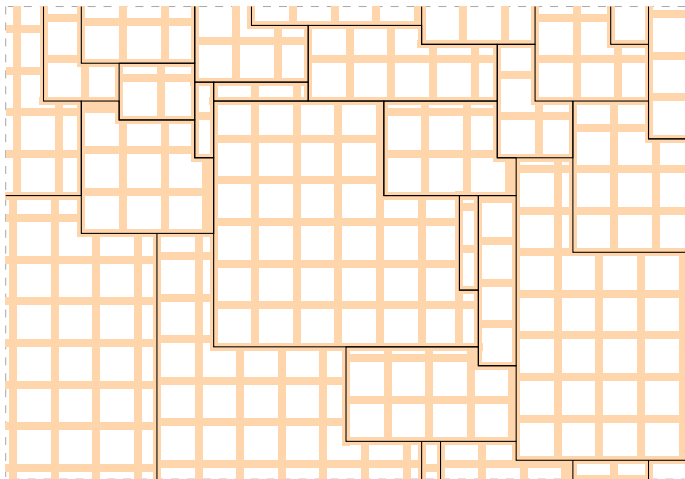
$$h(s/L + v/N) = (x, t) \text{ and } \tilde{h}(s/L + v/N) = (\tilde{x}, \tilde{t})$$

Forbidden regions I



- Regions I avoided $\Rightarrow x = \tilde{x}$

Forbidden regions II



- Regions II avoided $\Rightarrow$ not only $x = \tilde{x}$ but also $\lfloor t/N \rfloor = \lfloor \tilde{t}/N \rfloor$

Approximate Homomorphism $h_{\approx}$

- ▶ Let $\mathcal{V} = \{0, \dots, qN - 1\}^n$
- ▶ We can efficiently compute a function $h_{\approx} : \mathcal{V} \rightarrow X \times \mathbb{N}$ with

$$h_{\approx}(v) = h_{\approx}(v') \iff \frac{v - v'}{N} \approx \lambda \text{ for some } \lambda \in \Lambda$$

- ▶ There is a collision iff v and v' differ approximately by an integer multiple of a lattice vector of the period lattice Λ
- ▶ The preimage $\mathcal{M} = h_{\approx}^{-1}(v)$ has the form

$$\mathcal{M} = \{v' = v + N\lambda + \xi_{\lambda} \in \mathcal{V} \mid \lambda \in \Lambda, \xi_{\lambda} \in (-1, 1)^n\}$$

provided that $s/L + v/N$ is not in a forbidden region III

Pseudo-Lattice States

- ▶ Let Λ be a full-rank lattice in $\mathbb{R}^n$ and $\mathcal{V} = \{0, \dots, qN - 1\}^n$
- ▶ A uniform superposition in $\mathbb{C}^{\mathcal{V}}$ over a set of the form

$$\mathcal{M} = \{v' = v + N\lambda + \xi_\lambda \in \mathcal{V} \mid \lambda \in \Lambda, \quad \xi_\lambda \in (-1, 1)^n\}$$

is called a pseudo-lattice state with period lattice Λ

- ▶ We can efficiently prepare such pseudo-lattice states with constant probability by evaluating the approximate homomorphism $h_\approx$
- ▶ The offset v is **random** and **different** each time we prepare a new pseudo-lattice state
- ▶ The perturbations ξ_λ spoil the lattice structure

Quantum Algorithm

- ▶ Prepare a pseudo-lattice state $|\mathcal{M}\rangle$ in $\mathbb{C}^{\mathcal{V}}$ with random offset v , i.e., a uniform superposition over a set of the form

$$\mathcal{M} = \{v' = v + N\lambda + \xi_\lambda \in \mathcal{V} \mid \lambda \in \Lambda, \quad \xi_\lambda \in (-1, 1)^n\}$$

- ▶ Embed $|\mathcal{M}\rangle$ into the **larger** Hilbert space $\mathbb{C}^{\mathcal{W}}$ where

$$\mathcal{W} = \{0, \dots, 2nqN - 1\}^n$$

to deal with the perturbation effects caused by ξ_λ

- ▶ Apply the multidimensional QFT in $\mathbb{C}^{\mathcal{W}}$ to the “remove” the random offset v by using the shift-invariance property of the Fourier transform
- ▶ Measure to obtain outcome w
- ▶ The vector $w/(2nq)$ is an approximation of a lattice vector λ^* of the dual lattice Λ^* of Λ

Sampling Approximate Vectors of the Dual Lattice Λ^*

- ▶ Let Λ^* denote the dual lattice of Λ , i.e.,

$$\Lambda^* = \{\lambda^* \in \mathbb{R}^n \mid \langle \lambda^*, \lambda \rangle \in \mathbb{Z} \text{ for all } \lambda \in \Lambda\}$$

- ▶ We have to consider approximate vectors of Λ^* since Λ^* has non-integer coefficients
- ▶ For $\lambda^* = (\lambda_1^*, \dots, \lambda_n^*) \in \Lambda^*$, define the set of good approximations by

$$\mathcal{R}_{\lambda^*} = \{(w_1, \dots, w_n) \in \mathbb{Z}^n \mid w_k \in \{ \lfloor 2nq\lambda_k^* \rfloor, \lfloor 2nq\lambda_k^* \rfloor + 1 \} \}$$

- ▶ For all $w \in \mathcal{R}_{\lambda^*}$, we have the approximation inequality

$$\left\| \frac{w}{2qn} - \lambda^* \right\|_2 \leq \frac{1}{2\sqrt{nq}}$$

Probability of Sampling an Approximate Vector of Λ^*

- ▶ For all $\lambda^* \in \Lambda^*$ with $\mathcal{R}_{\lambda^*} \subset [0, 2qn\kappa N]^n$ for some fixed $\kappa \in (0, 1)$, we obtain the improved lower bound

$$\Pr(\mathcal{R}_{\lambda^*}) \gtrsim 2^{n-1} \cdot \frac{1}{(2qnN)^n} \cdot \frac{q^n}{\det(\Lambda)} \cdot \cos^2\left(\pi\left(\frac{1}{4} + \frac{1}{2qN} + 2\kappa n\right)\right)$$

- ▶ The number of such $\lambda^* \approx \frac{(\kappa N)^n}{\det(\Lambda^*)} = (\kappa N)^n \det(\Lambda)$
- ▶ The probability of obtaining an approximation of an essentially uniformly selected lattice vector $\lambda^* \in \Lambda^* \cap [0, \kappa N]^n$ is

$$p_1 \gtrsim \frac{c}{2} \left(\frac{\kappa}{n}\right)^n$$

- ▶ The restriction to $[0, 2qn\kappa N]^n$ with $\kappa < 1$ allows us to deal with the perturbations effects caused by the deviations $(w/2qn) - \lambda^*$ for $w \in \mathcal{R}_{\lambda^*}$

Probability of Generating a Full-Rank Lattice L

- ▶ Let L be a full-rank lattice in $\mathbb{R}^n$
- ▶ Assume $b_1, \dots, b_n$ chosen uniformly at random in $L \cap [0, B)^n$
- ▶ With some constant probability these vectors generate a full-rank sublattice L_0
- ▶ L/L_0 is a finite abelian group that can be generated with only n elements
- ▶ Assume $b_{n+1}, \dots, b_{2n+1}$ are chosen uniformly at random in $L \cap [0, B_0)^n$ with $B_0 > B$
- ▶ The elements $b_{n+k} + L_0$ are nearly uniformly distributed over L/L_0 for $k = 1, \dots, n+1$
- ▶ With some constant probability p the samples $b_1 \dots, b_{2n+1}$ generate L

Approximate Basis for the Period Lattice Λ (finally!)

- ▶ Repeating the quantum algorithm $2n + 1$ times yields the approximate generating set

$$\frac{w_1}{2nq}, \dots, \frac{w_{2n+1}}{2nq}$$

of Λ^* with probability at least

$$p p^{2n+1} = p \left(\frac{c \kappa^n}{2n^n} \right)^{2n+1}$$

where $c = \cos^2 \left(\pi \left(\frac{1}{4} + \frac{1}{2qN} + 2\kappa n \right) \right)$ and $\kappa \geq \frac{1}{8n}$

- ▶ We cover an approximate basis of Λ from the above approximate generating set by a classical algorithm
- ▶ We recover an approximate basis of Λ from the approximate basis of Λ^* by matrix inversion

Improvements

- ▶ We present a rigorous and complete proof that the running time is polynomial in the determinant of the period lattice Λ
- ▶ We give an explicit lower bound on the success probability

HALLGREN and SCHMIDT AND VOLLMER did not give any lower bound

SCHMIDT presented a lower bound in his dissertation which is worse than our bound

- ▶ We improve the success probability by at least 2^{n^2-1}
- ▶ But the resulting lower bound is still quite bad $\Rightarrow$ find new ideas and techniques to improve it

Thank you!!!